OPIM101 Decision Analysis

Lecture 12

Lecture 9:
Decision Making under Risk and Uncertainty
Probability analysis
Bayes Theorem

Lecture 10:
Decision Making under Risk and Uncertainty
Decision Rules and Sensitivity Analysis

Lecture 11:
Decision Making under Risk and Uncertainty
Decision Trees and Value of Information

Lecture 12:
Simulation and Applications
Computer Implementation (Excel)

Outline

Decision Making under Uncertainty Using Simulation

- What is simulation
- The Monte Carlo Simulation
- Simulation with Excel Spreadsheets
- Areas of Simulation Application

Guessing…

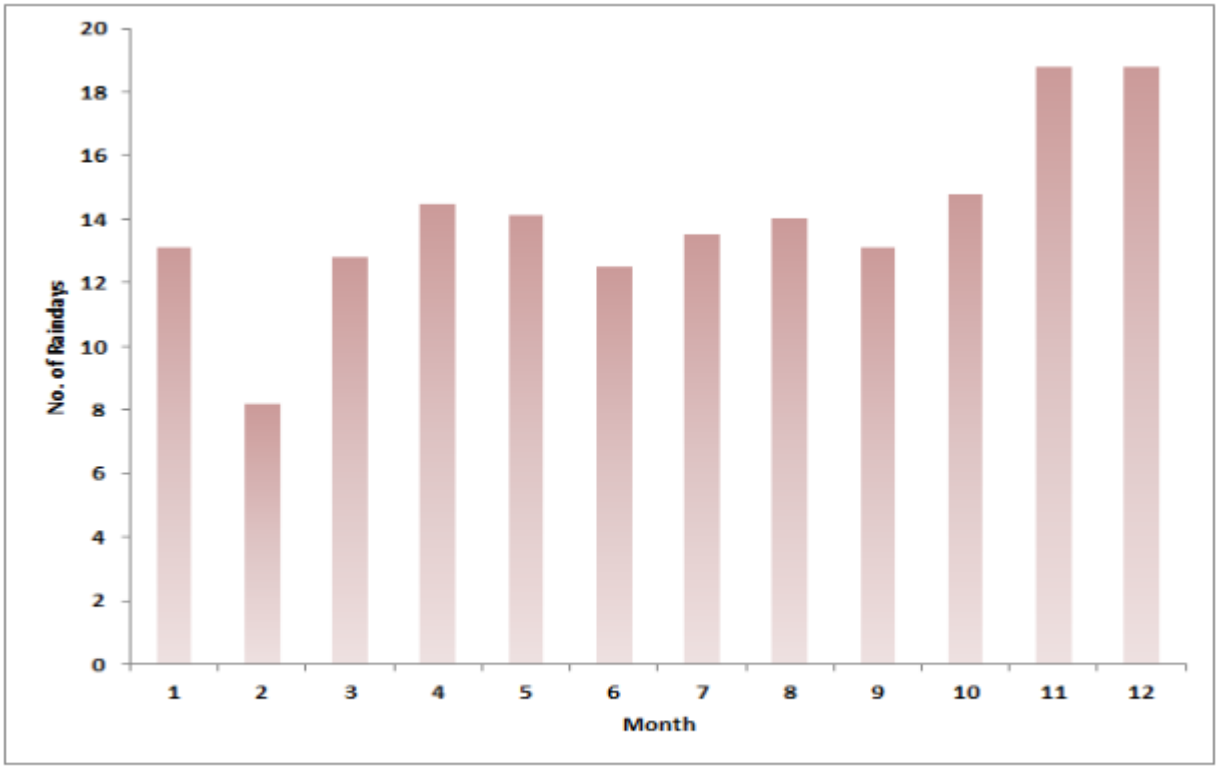


Fig 1 – Average number of rain days per month (1982-2016).

** Data from Changi Climate Station*

How many rainy days will there be in next month?

Simulation in Systems Engineering

When dealing with complex or uncertain systems, simulations can be used to model the probability distribution of different outcomes, and to estimate the risks and uncertainties associated with a given decision.

- E.g., How to best design the layout and processes of a container yard?

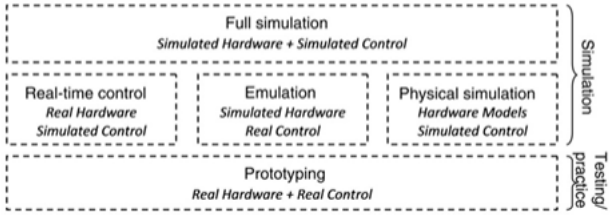
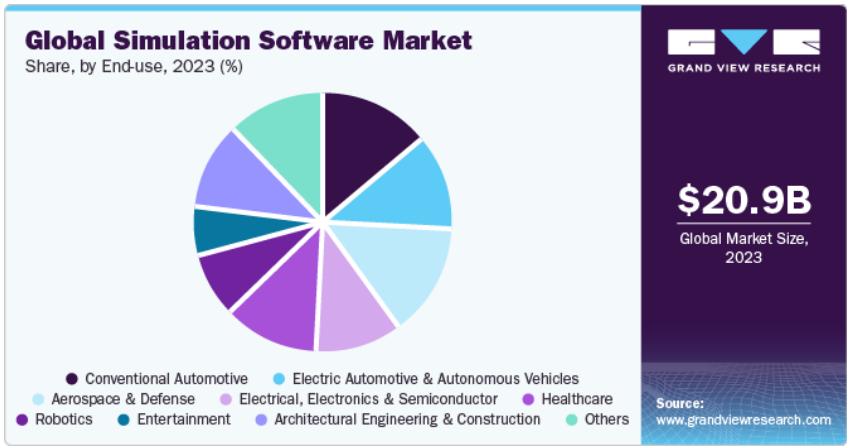


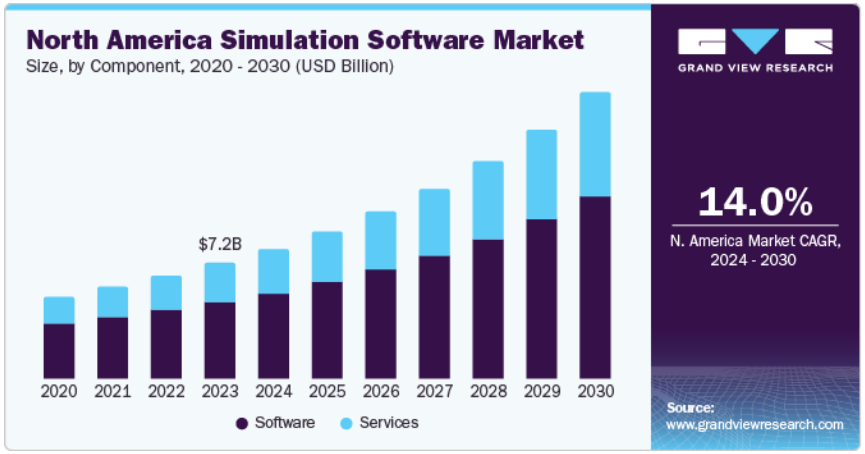
Figure 2. Classification of simulation models according to their reliance to real port components.

Port terminal simulation:
https://www.youtube.com/watch?v=fQZ6lLiZ3aw

Angeloudis P, Bell MGH. A review of container terminal simulation models. *Maritime Policy & Management*. 2011;38(5):523-540. doi:10.1080/03088839.2011.597448

Simulation Software





Overview on Simulation

Advantages	Disadvantages
• Advances in technology makes simulation viable. • It can be used to analyse large and complex real-world situations that cannot be solved "exactly". • It can be used to study "What-if?" scenarios • It can simulate the problem without running the actual experiments which will interfere the real-world system.	• Good simulation models for complex problems can be expensive. • Simulation alone does not gives optimal solutions. • Each simulation model is unique, hence it cannot be transferrable to other problems.

Monte Carlo Process

- A large proportion of the applications of simulations are for ***probabilistic models***.

- The ***Monte Carlo*** technique is defined as a technique for selecting numbers ***randomly*** from a probability distribution for use in a *trial* (computer run) of a simulation model.

- The basic principle behind the process is the same as in the operation of gambling devices in casinos (such as those in Monte Carlo, Monaco).

Monte Carlo Process
Use of Random Numbers

- In the Monte Carlo process, values for a random variable are generated by sampling from a probability distribution.
- Example: ComputerWorld demand data for laptops selling for $4,300 over a period of 100 weeks.

PCs Demanded per Week	Frequency of Demand	Probability of Demand, $P(x)$
0	20	.20
1	40	.40
2	20	.20
3	10	.10
4	10	.10
	100	1.00

Table 14.1 Probability distribution of demand for laptop PCs

Monte Carlo Process
Use of Random Numbers

- The purpose of the ***Monte Carlo process is to generate the random variable,*** demand, by sampling from the probability distribution P(x).

- The partitioned roulette wheel replicates the probability distribution for demand if the values of demand occur in a random manner.

- The segment at which the wheel stops indicates demand for one week.

Monte Carlo Process
Use of Random Numbers

PCs Demanded per Week	Frequency of Demand	Probability of Demand, $P(x)$
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Table 14.1 Probability distribution of demand for laptop PCs

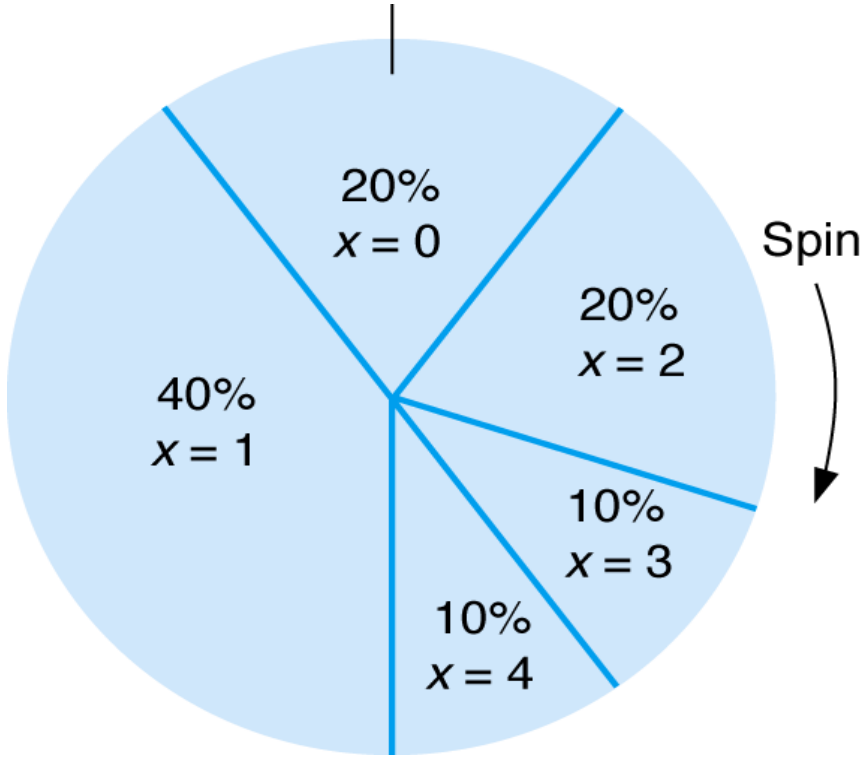
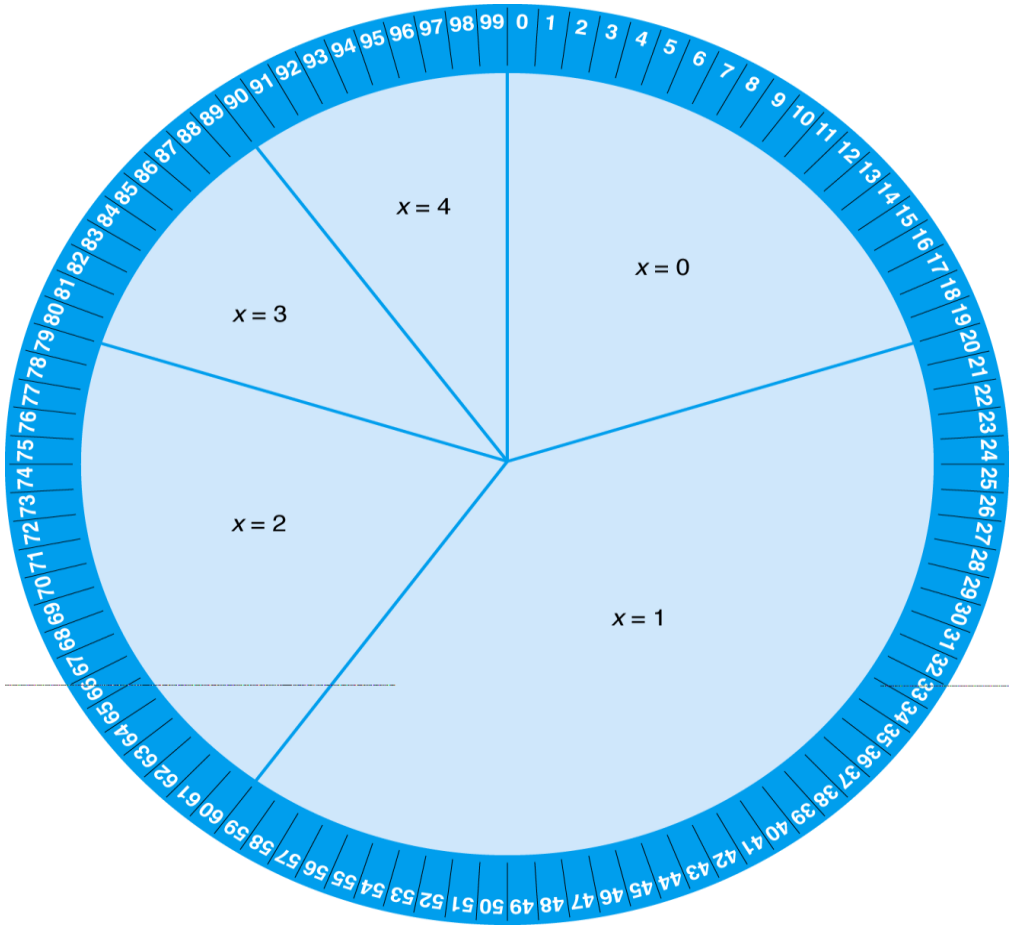


Figure 14.1 A roulette wheel for demand

Monte Carlo Process
Use of Random Numbers

When the wheel is spun, the actual demand for PCs is determined by a number at rim of the wheel.



Monte Carlo Process
Use of Random Numbers

Demand, x	Ranges of Random Numbers, r
0	0–19
1	20–59
2	60–79
3	80–89
4	90–99

$r = 39$

Table 14.2 Generating demand from random numbers

Monte Carlo Process
Use of Random Numbers

- Repeating the selection of random numbers simulates demand for a period of time. The table shows demand for 15 consecutive weeks as drawn from the random number table.
- Estimated average demand = 31/15 = 2.07 laptop PCs per week.
- Estimated average revenue = $133,300/15 = $8,886.67.

Week	r	Demand, x	Revenue ($)
1	39	1	4,300
2	73	2	8,600
3	72	2	8,600
4	75	2	8,600
5	37	1	4,300
6	02	0	0
7	87	3	12,900
8	98	4	17,200
9	10	0	0
10	47	1	4,300
11	93	4	17,200
12	21	1	4,300
13	95	4	17,200
14	97	4	17,200
15	69	2	8,600
		$\Sigma = 31$	$133,300

Monte Carlo Process
Use of Random Numbers

Average demand could have been calculated *analytically:*

$$E(x) = \sum_{i=1}^{n} P(x_i)x_i$$

where:

$x_i =$ demand value i
$P(x_i) =$ probability of demand
$n =$ the number of different demand values

therefore:

$$E(x) = (.20)(0) + (.40)(1) + (.20)(2) + (.10)(3) + (.10)(4)$$
$$= 1.5 \text{ PCs per week}$$

Monte Carlo Process
Use of Random Numbers

- The ***more periods*** simulated, the ***more accurate*** the results.
- Simulation results will <u>not</u> equal analytical results unless enough trials have been conducted to reach ***steady state***.
- It is often difficult to ***validate*** results of simulation - that true steady state has been reached and that simulation model truly replicates reality.
- <u>When analytical analysis is not possible</u>, there is no analytical standard of comparison, thus making validation even more difficult.

Seeing the "Expectation"

distribution. More precisely, it is defined as the probability-weighted sum of all possible values in the random variable's support,

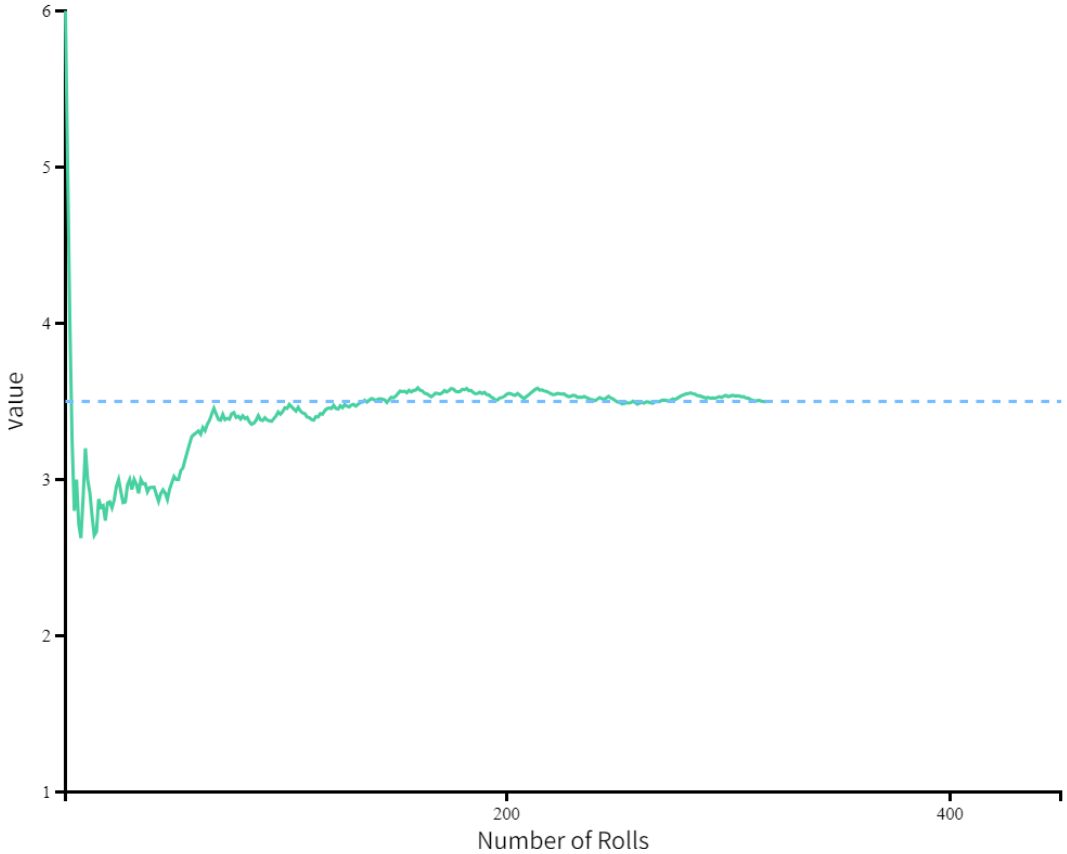
$$\mathrm{E}[X] = \sum_{x \in \mathcal{X}} x P(x)$$

Consider the probabilistic experiment of rolling a fair die and watch as the running sample mean converges to the expectation of 3.5.

Roll the Die

Roll 100 times

Change the distribution of the different faces of the die (thus making the die biased or "unfair") by adjusting the blue bars below and observe how this changes the expectation.



https://seeing-theory.brown.edu/basic-probability/index.html

Computer Simulation with Excel Spreadsheets Generating Random Numbers

- As simulation models get ***more complex*** they become ***impossible to perform manually.***

- In simulation modeling, random ***numbers are generated by a mathematical process*** instead of a physical process (such as wheel spinning).

- Random numbers are typically generated on the computer using a numerical technique and thus are not true random numbers but ***pseudorandom numbers***.

Monte Carlo Process
Use of Random Numbers

Select a number from a random number table:

97 18 31 55 73	10 65 81 92 59	77 31 61 95 46	20 44 90 32 64	26 99 76 75 63
69 08 88 86 13	59 71 74 17 32	48 38 75 93 29	73 37 32 04 05	60 82 29 20 25
41 26 10 25 03	87 63 93 95 17	81 83 83 04 49	77 45 85 50 51	79 88 01 97 30
91 47 14 63 62	08 61 74 51 69	92 79 43 89 79	29 18 94 51 23	14 85 11 47 23
80 94 54 18 47	08 52 85 08 40	48 40 35 94 22	72 65 71 08 86	50 03 42 99 36
67 06 77 63 99	89 85 84 46 06	64 71 06 21 66	89 37 20 70 01	61 65 70 22 12
59 72 24 13 75	42 29 72 23 19	06 94 76 10 08	81 30 15 39 14	81 33 17 16 33
63 62 06 34 41	79 53 36 02 95	94 61 09 43 62	20 21 14 68 86	84 95 48 46 45
78 47 23 53 90	79 93 96 38 63	34 85 52 05 09	85 43 01 72 73	14 93 87 81 40
87 68 62 15 43	97 48 72 66 48	53 16 71 13 81	59 97 50 99 52	24 62 20 42 31
47 60 92 10 77	26 97 05 73 51	88 46 38 03 58	72 68 49 29 31	75 70 16 08 24
56 88 87 59 41	06 87 37 78 48	65 88 69 58 39	88 02 84 27 83	85 81 56 39 38
22 17 68 65 84	87 02 22 57 51	68 69 80 95 44	11 29 01 95 80	49 34 35 36 47
19 36 27 59 46	39 77 32 77 09	79 57 92 36 59	89 74 39 82 15	08 58 94 34 74
16 77 23 02 77	28 06 24 25 93	22 45 44 84 11	87 80 61 65 31	09 71 91 74 25
78 43 76 71 61	97 67 63 99 61	30 45 67 93 82	59 73 19 85 23	53 33 65 97 21
03 28 28 26 08	69 30 16 09 05	53 58 47 70 93	66 56 45 65 79	45 56 20 19 47
04 31 17 21 56	33 73 99 19 87	26 72 39 27 67	53 77 57 68 93	6
61 06 98 03 91	87 14 77 43 96	43 00 65 98 50	45 60 33 01 07	9
23 68 35 26 00	99 53 93 61 28	52 70 05 48 34	56 65 05 61 86	9
15 39 25 70 99	93 86 52 77 65	15 33 59 05 28	22 87 26 07 47	8
58 71 96 30 24	18 46 23 34 27	85 13 99 24 44	49 18 09 79 49	7
93 22 53 64 39	07 10 63 76 35	87 03 04 79 88	08 13 13 85 51	5
78 76 58 54 74	92 38 70 96 92	52 06 79 79 45	82 63 18 27 44	6
61 81 31 96 82	00 57 25 60 59	46 72 60 18 77	55 66 12 62 11	0
42 88 07 10 05	24 98 65 63 21	47 21 61 88 32	27 80 30 21 60	1
77 94 30 05 39	28 10 99 00 27	12 73 73 99 12	49 99 57 94 82	96 88 57 17 91

Table 14.3 A table of delightfully random numbers

In **1956**, the RAND corporation published the book: *A Million Random Digits with 100,000 Normal Deviates.*

Computer Simulation with Excel Spreadsheets
Generating Random Numbers

Artificially created random numbers must have the following characteristics:

1. The random numbers must be ***uniformly distributed***.
2. The numerical technique for generating the numbers must be ***efficient***.
3. The sequence of random numbers should ***reflect no pattern***.

Illustration with Excel

- In the Monte Carlo process, values for a random variable are generated by sampling from a probability distribution.
- Example: ComputerWorld has the following demand data for PCs.

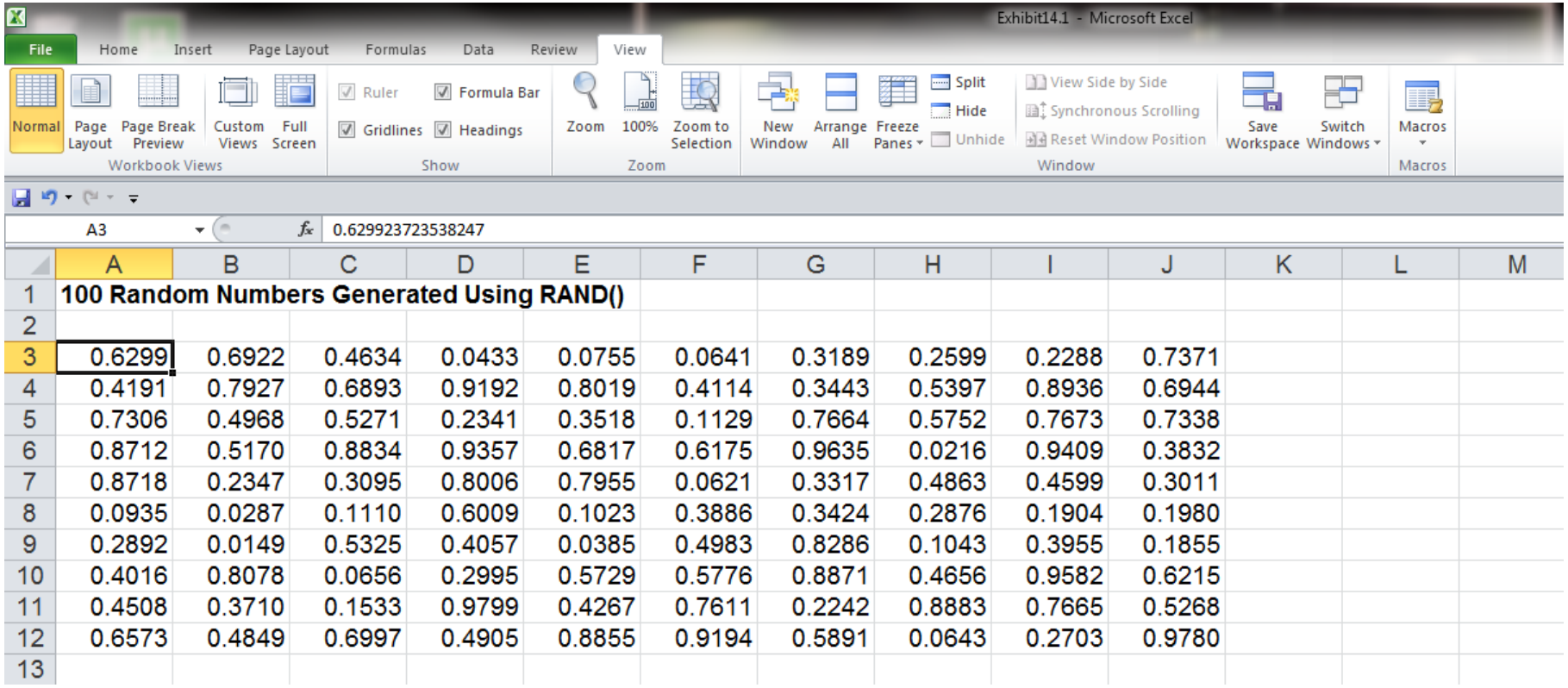
PCs Demanded per Week	Frequency of Demand	Probability of Demand, $P(x)$
0	20	.20
1	40	.40
2	20	.20
3	10	.10
4	10	.10
	100	1.00

Using simulation, what is the demand of PC for the next 10 weeks?

Generate Random number with Excel

Using "Fixed" random numbers

Simulation with Excel Spreadsheets (1 of 2)
Uniform between 0 -1



	A	B	C	D	E	F	G	H	I	J
1	100 Random Numbers Generated Using RAND()									
2										
3	0.6299	0.6922	0.4634	0.0433	0.0755	0.0641	0.3189	0.2599	0.2288	0.7371
4	0.4191	0.7927	0.6893	0.9192	0.8019	0.4114	0.3443	0.5397	0.8936	0.6944
5	0.7306	0.4968	0.5271	0.2341	0.3518	0.1129	0.7664	0.5752	0.7673	0.7338
6	0.8712	0.5170	0.8834	0.9357	0.6817	0.6175	0.9635	0.0216	0.9409	0.3832
7	0.8718	0.2347	0.3095	0.8006	0.7955	0.0621	0.3317	0.4863	0.4599	0.3011
8	0.0935	0.0287	0.1110	0.6009	0.1023	0.3886	0.3424	0.2876	0.1904	0.1980
9	0.2892	0.0149	0.5325	0.4057	0.0385	0.4983	0.8286	0.1043	0.3955	0.1855
10	0.4016	0.8078	0.0656	0.2995	0.5729	0.5776	0.8871	0.4656	0.9582	0.6215
11	0.4508	0.3710	0.1533	0.9799	0.4267	0.7611	0.2242	0.8883	0.7665	0.5268
12	0.6573	0.4849	0.6997	0.4905	0.8855	0.9194	0.5891	0.0643	0.2703	0.9780
13										

Simulation with Excel Spreadsheets (2 of 2)
Discrete Random Number

=VLOOKUP(F6,Lookup,2)

	A	B	C	D	E	F	G
1	**ComputerWorld Simulation Example**						
2							
3	*Probability of Weekly Demand:*				*Simulation:*		
4							
5	P(x)	Cumulative	Demand		Week	RN	Demand
6	0.20	0	0		1	0.6299	2
7	0.40	0.20	1		2	0.4191	1
8	0.20	0.60	2		3	0.7306	2
9	0.10	0.80	3		4	0.8712	3
10	0.10	0.90	4		5	0.8718	3
11	1.00				6	0.0935	0
12					7	0.2892	1
13	*Average Demand =*		1.60		8	0.4016	1
14					9	0.4508	1
15					10	0.6573	2
16						*Total*	16
17							

To generate distribution

- Uniform discrete: randbetween(bottom, top)
  - Returns a random integer number between bottom and top (inclusive).

- If it is normal distribution, we use NORM.DIST

Statistical Analysis

Outcomes of Simulation model

- We often use the average as the outcome of the simulation model. Is this the best we can do?
- The outcome of the simulation model is a random variable by itself
- We should present the statistical results, rather than the average

Source: The Flaw of Averages by Sam L. Savage, Hoboken, NJ: John Wiley & Sons, 2009.

The Flaw of Averages

Using Confidence Interval

- Suppose the simulation model states that the average is $\bar{x}$

- Then, we can be 95% confident that the true value lies in the interval

$$\left(\bar{x} - 1.96\left(\frac{s}{\sqrt{n}}\right), \bar{x} + 1.96\left(\frac{s}{\sqrt{n}}\right)\right)$$

where s is the sample standard deviation, and n is the number of trials performed in the simulation

Evaluation by simulation

Illustration with Excel

- In the Monte Carlo process, values for a random variable are generated by sampling from a probability distribution.
- Example: ComputerWorld has the following demand data for PCs.

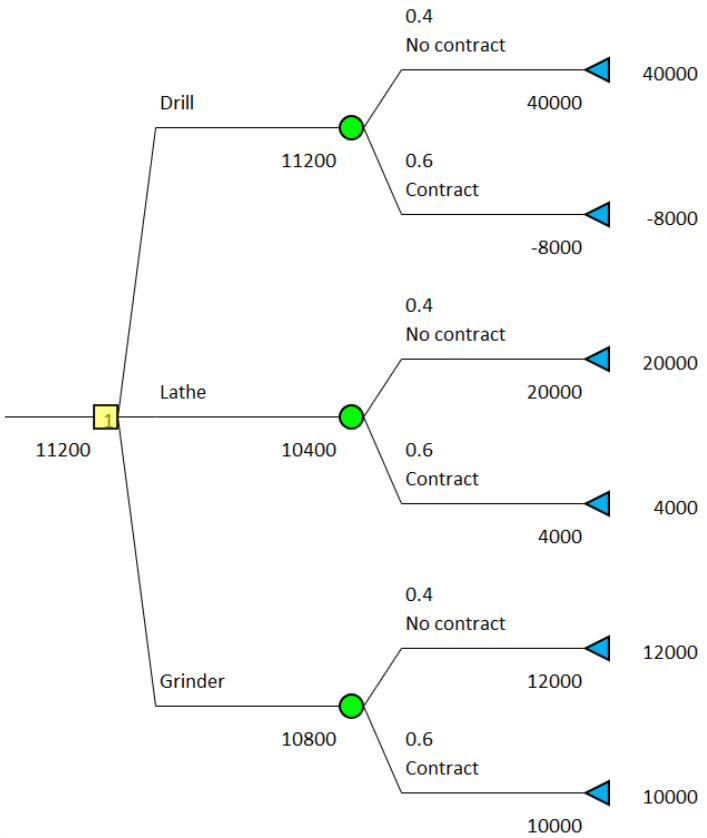
PCs Demanded per Week	Frequency of Demand	Probability of Demand, $P(x)$
0	20	.20
1	40	.40
2	20	.20
3	10	.10
4	10	.10
	100	1.00

What is the 95% confidence interval for the average demand?

Decision Tree Practice with Simulation

- A machine shop owner is attempting to decide whether to purchase a new drill press, a lathe or a grinder. The return from each will be determined by whether the company succeeds in getting a government military contract. The profit or loss from each purchase and the probabilities associated with each contract outcome are shown in the following payoff table:

Purchase	States of Nature	
	Contract **0.4**	No Contract **0.6**
Drill press	$ 40,000	$ -8,000
Lathe	20,000	4,000
Grinder	12,000	10,000



a) Use the expected value to decide on the purchase → Choose Drill press

Question: How does this perform in practice?

Payoff Table Example and Simulation

- Palm Garden Greenhouse grows carnations & sell to florists
  - Carnations sold for $3 per dozen, cost $2 per dozen.

a) Develop payoff table. Consider the decisions to produce 20, 22, 24, 26, 28 and 30.

b) Compute the expected value for each decision and select the best decision.

c) How is the best policy performing under simulation? Compare the performance of the two best decisions.

Daily Demand	Probability
20	0.05
22	0.10
24	0.25
26	0.30
28	.20
30	0.10
Total	1.00

Process simulation

Practice M1: First step in simulation

Maria runs a bakery shop specializing in birthday cakes. Based on the past sales, she constructed the following demand distribution:

Demand (cakes/day)	Probability
5	0.05
10	0.20
15	0.40
20	0.20
25	0.15

1) Simulate the demand for the next 10 days and find the average demand.

2) Find the average demand in part 1)

3) She bakes 20 cakes a day. Each cake costs her $3. She sells a cake for $10. Unsold cakes are discarded. Using simulation, what is the profit for the next 10 days?

2 random variables simulation problem

Practice M3: 2 random variables

Maria runs a bakery shop specializing in birthday cakes. Based on the past sales, she constructed the following demand distribution:

Demand (cakes/day)	Probability
5	0.05
10	0.20
15	0.40
20	0.20
25	0.15

No. of Bargain hunters	Probability
5	0.45
10	0.30
15	0.25

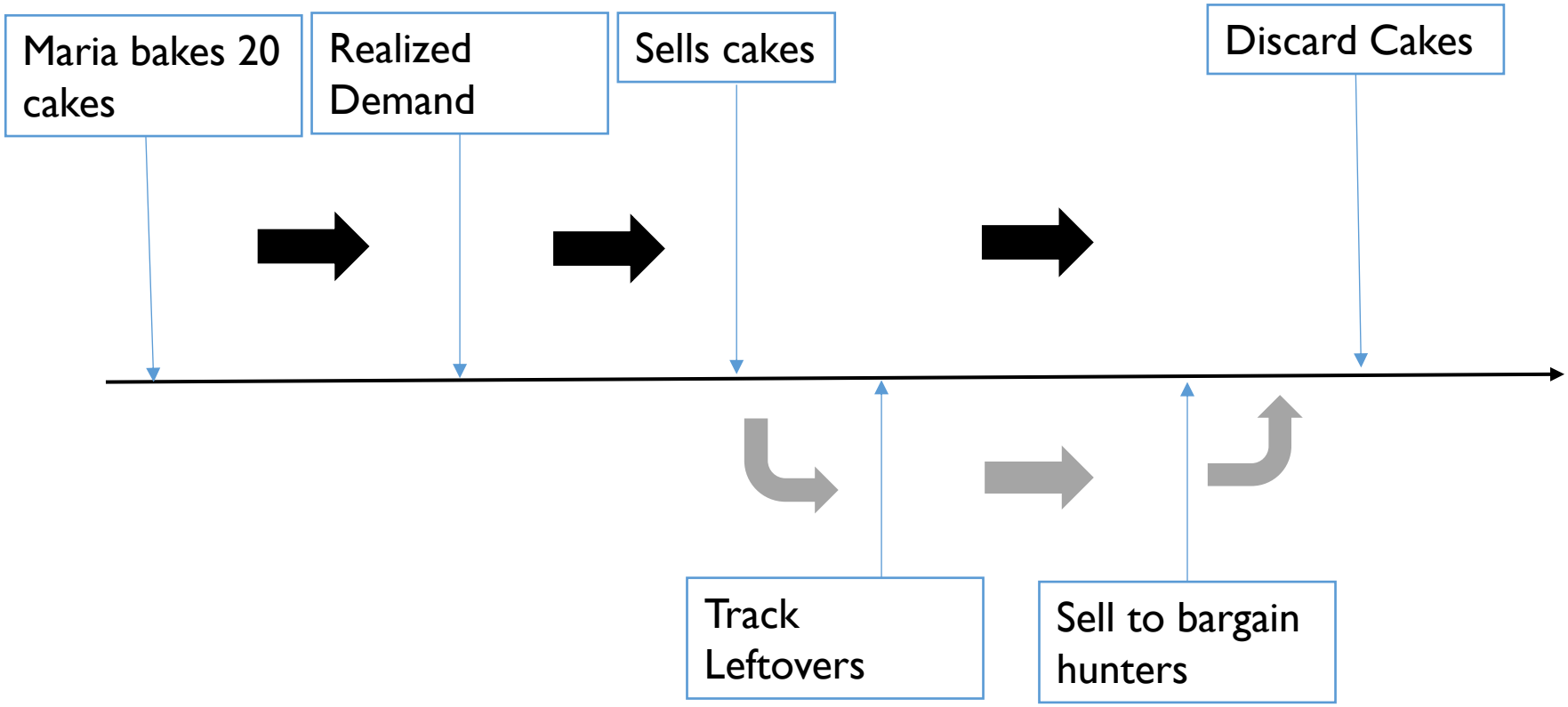
1) Simulate the demand for the next 10 days and find the average demand.

2) Use the second column of the random numbers and find the average demand in part 1).

3) She bakes 20 cakes a day. Each cake costs her $3. She sells a cake for $10. **Unsold cakes are discarded**. Using simulation, what is the profit for the next 10 days?

4) Maria does not like to see wastages. **Before discarding**, she decides to sell cakes which are unsold at a **discounted rate (50% off) to bargain hunters**. What is the profit for the next 10 days?

Sequence of events



Process Simulation: Queuing Problem

- Raw materials arrive and are processed in the factory
- The arrival times and processing times are both random

Application: Queuing Problem

Arrival Interval (days), x	Probability, $P(x)$	Cumulative Probability	Random Number Range, r_1
1.0	.20	.20	1–20
2.0	.40	.60	21–60
3.0	.30	.90	61–90
4.0	.10	1.00	91–99, 00

Distribution of arrival intervals

Service Time (days), y	Probability, $P(y)$	Cumulative Probability	Random Number Range, r_2
0.5	.20	.20	1–20
1.0	.50	.70	21–70
2.0	.30	1.00	71–99, 00

Distribution of service times

Application: Queuing Problem

i. Map out the sequence of events before working on the simulation
ii. Create 2 random variables and map out the realized values
iii. Link the random variables with the events

Batch	r_1	Arrival Interval, x	Arrival Clock	Enter Facility Clock	Waiting Time	r_2	Service Time, y	Departure Clock	Time in System
1	—	—	0.0	0.0	0.0	65	1.0	1.0	1.0
2	71	3.0	3.0	3.0	0.0	18	0.5	3.5	0.5
3	12	1.0	4.0	4.0	0.0	17	0.5	4.5	0.5
4	48	2.0	6.0	6.0	0.0	89	2.0	8.0	2.0
5	18	1.0	7.0	8.0	1.0	83	2.0	10.0	3.0
6	08	1.0	8.0	10.0	2.0	90	2.0	12.0	4.0
7	05	1.0	9.0	12.0	3.0	89	2.0	14.0	5.0
8	18	1.0	10.0	14.0	4.0	08	0.5	14.5	4.5
9	26	2.0	12.0	14.5	2.5	47	1.0	15.5	3.5
10	94	4.0	16.0	16.0	0.0	06	0.5	16.5	0.5
					12.5				**24.5**

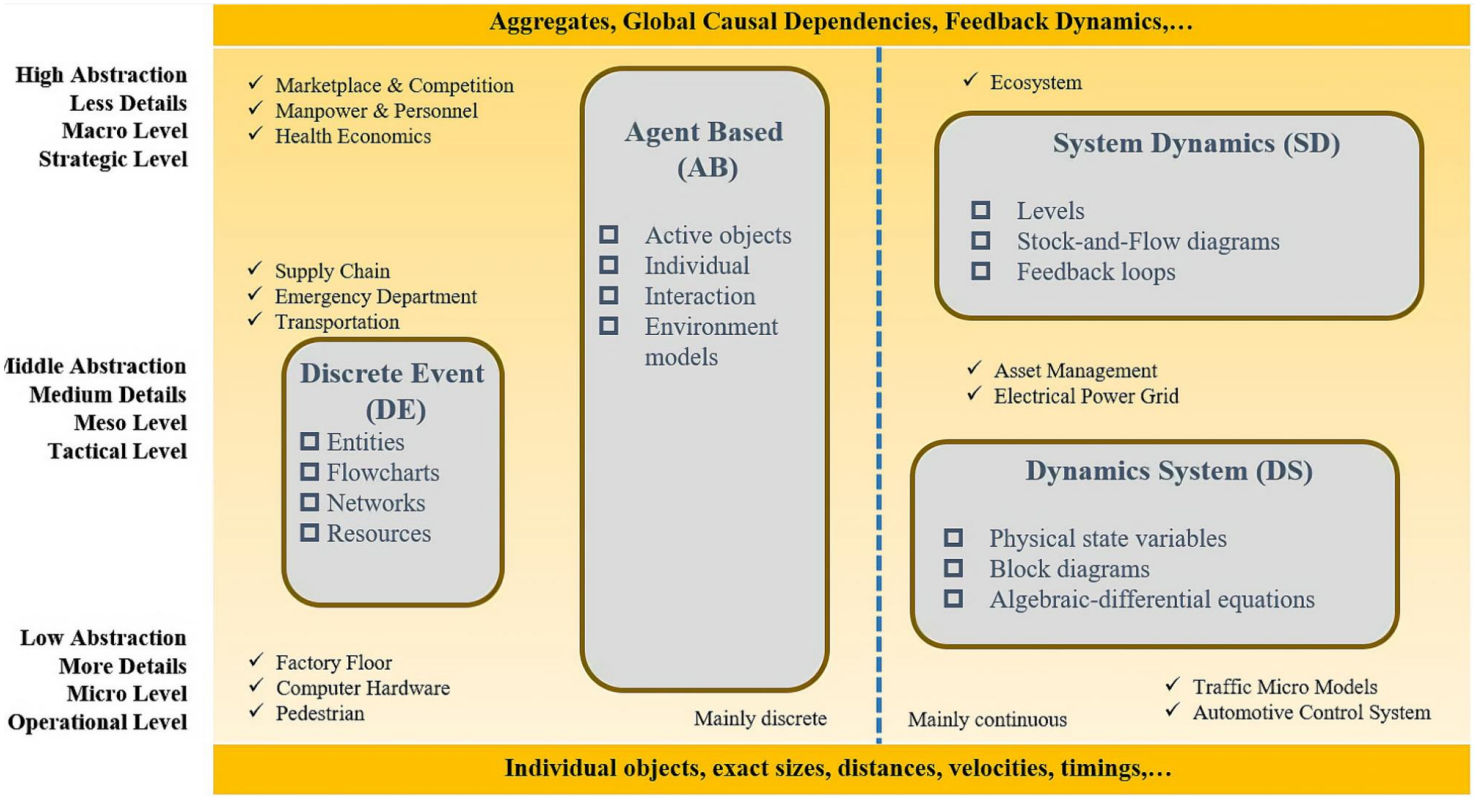
Validity of the Simulation Model (1 of 2)

- The analyst wants to be certain that the *model is internally correct* and that all operations are *logical and mathematically correct*.
- Testing procedures for validity:
  - Run a small number of trials of the model and compare with *manually derived solutions*.
  - *Divide the model* into parts and run parts separately to reduce the complexity of checking.
  - *Simplify mathematical relationships* (if possible) for easier testing.
  - *Compare results* with actual real-world data.

Validity of the Simulation Model (2 of 2)

- The analyst must determine if the model's starting conditions are correct (system empty, etc.).
- Must determine how long model should run to insure steady-state conditions.
- A standard, fool-proof procedure for validation is not available.
- Validity of the model rests ultimately on the expertise and experience of the model developer.
- Process is sometimes known as ***Verification and Validation (V&V)***

Simulation Modelling Paradigms



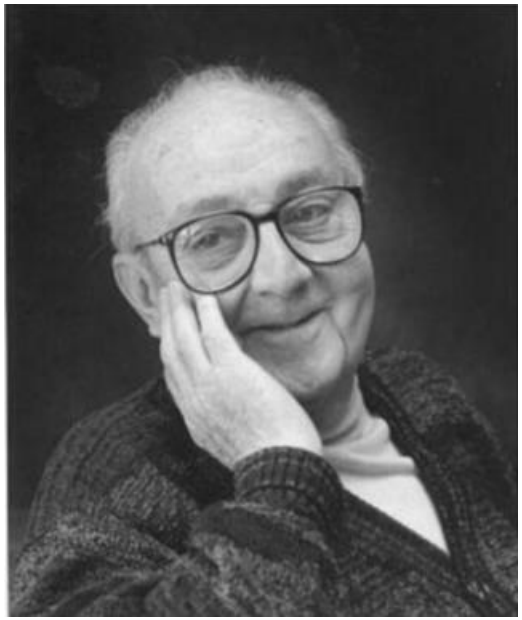
Qiu H, Chen Y, Zhang H, Yi W, Li Y. Evolutionary digital twin model with an agent-based discrete-event simulation method. *Appl Intell*. 2023;53(6):6178-6194. doi:10.1007/s10489-022-03507-2

Needs-driven Approach

"The most that can be expected from any model is that it can supply a **useful** approximation to reality:

All models are wrong; some models are useful."

George E. P. Box (1919-2013)
One of the great statistical minds of the 20th century

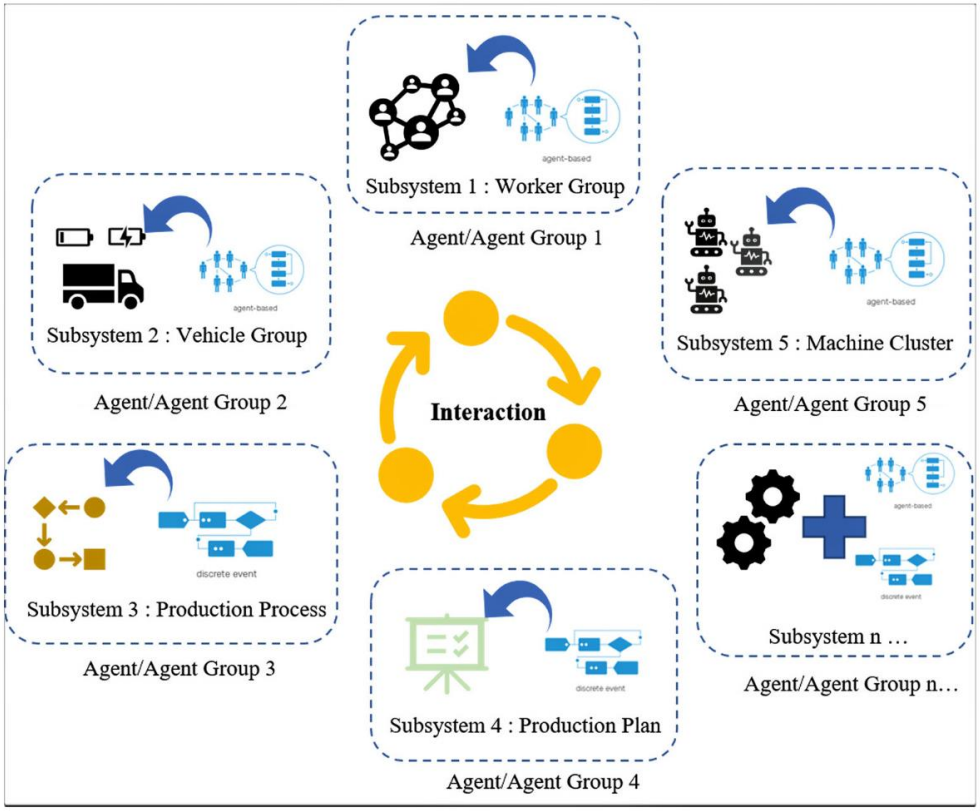


Some Areas of Simulation Application

- Queuing
- Inventory Control
- Production and Manufacturing
- Finance
- Marketing
- Public Service Operations
- Environmental and Resource Analysis

Digital Twins (DT)

- "DT is an integrated multiphysics, multiscale, probabilistic simulation of a vehicle or system that uses the best available physical models, sensor updates, fleet history, and so forth, to mirror the life of its flying twin."



Summary of Lecture 12

- Monte Carlo simulation with Excel
  - Generate random variables
- Perform "What if" analysis
  - Use simulation for making decision
- Applications